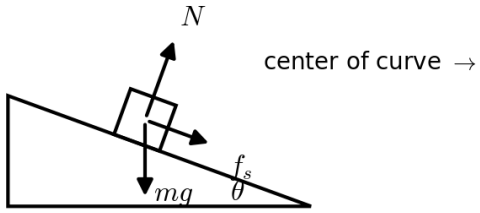
# PHYS1408 Practice Exam 2-3

**Chapters 6, 7/8, 9, 10, 11**

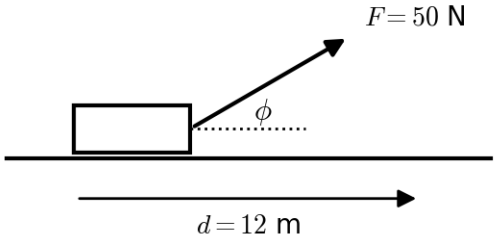
Use $g = 9.8$ m/s$^2$. Choose the single best answer. Answers are rounded to the precision shown. For a minimum or maximum, choose the answer choice that actually works.

**1.** A highway curve of radius 78 m is banked at $\theta = 20°$. The coefficient of static friction between the tires and the road is $\mu_s = 0.25$. What is the maximum speed at which a car can take the curve without sliding up the bank?
a) 9 m/s b) 14 m/s c) 17 m/s d) 21 m/s e) 23 m/s

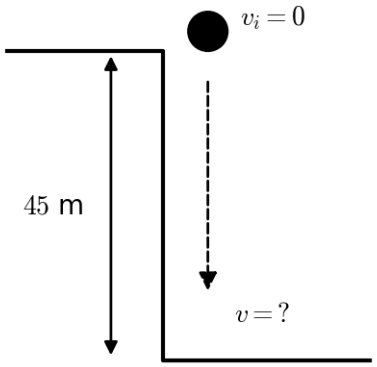



**2.** As defined in class, energy is
a) the rate at which work is done.
b) a vector quantity measured in newtons.
c) the product of mass and velocity.
d) the resistance of an object to rotation.
e) the ability to do work.

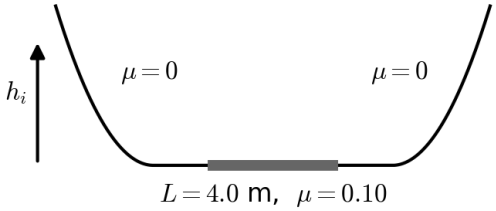
**3.** A sled is pulled 12 m across level snow by a rope that makes an angle of 30° above the horizontal. The tension in the rope is 50 N. How much work does the rope do on the sled?
a) 0 J b) 300 J c) 520 J d) 600 J e) 1200 J



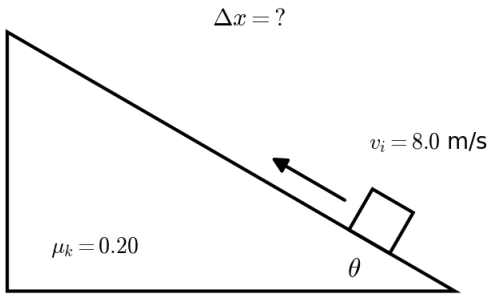
**4.** A coyote steps off the edge of a 45 m cliff, starting from rest. Ignoring air resistance, how fast is it moving just before it hits the ground below?
a) 29.7 m/s b) 42.0 m/s c) 59.4 m/s d) 441 m/s
e) 882 m/s



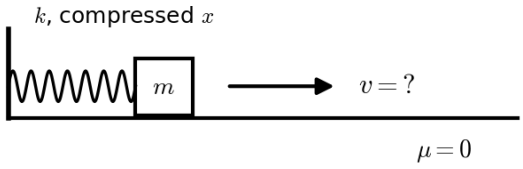

**5.** A skateboarder starts from rest 1.8 m up one wall of a halfpipe. The halfpipe is frictionless except for a 4.0 m long section of the flat bottom with $\mu = 0.10$. She rolls back and forth until she comes to rest. How many times does she cross the **entire** rough section?
a) 1 b) 2 c) 3 d) 4 e) 5



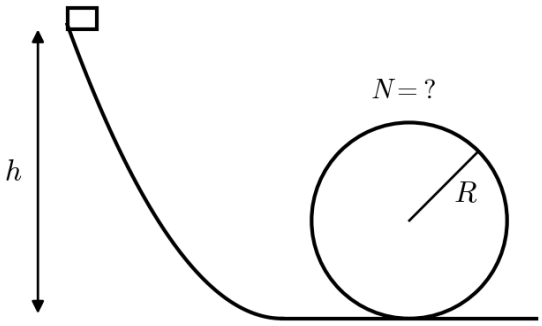

**6.** A 2.0 kg block slides **up** a 30° incline ($\mu_k = 0.20$) with an initial speed of 8.0 m/s. How far along the incline does it slide before it comes to rest?
a) 0.61 m b) 2.42 m c) 3.27 m d) 4.66 m e) 4.85 m



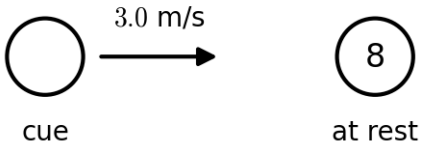

**7.** A horizontal spring ($k = 800$ N/m) is compressed 0.10 m. A 0.20 kg block placed against it is released and slides along a frictionless floor. How fast is the block moving after it leaves the spring?
a) 4.47 m/s b) 6.32 m/s c) 8.94 m/s d) 20.0 m/s
e) 40.0 m/s



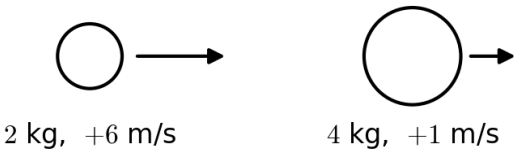

**8.** A 500 kg roller coaster car starts from rest at a height $h = 30$ m and slides down a frictionless track into a vertical circular loop of radius $R = 10$ m. What is the normal force the track exerts on the car at the **top** of the loop?
a) 0 N b) 4900 N c) 9800 N d) 14,700 N e) 24,500 N



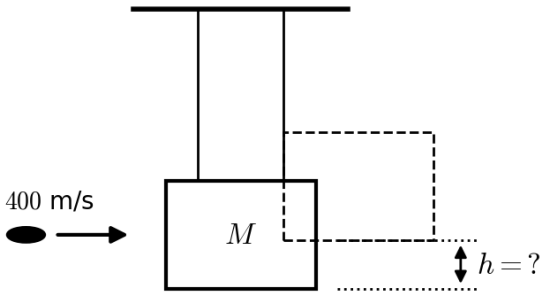
**9.** A cue ball moving at 3.0 m/s hits an 8-ball of the same mass head on. The 8-ball is initially at rest, and the collision is elastic. What is the velocity of the cue ball right after the collision? (Positive is the cue ball's original direction.)
a) −3.0 m/s b) −1.5 m/s c) 0 m/s d) +1.5 m/s
e) +3.0 m/s



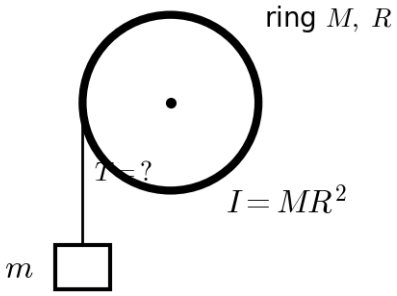

**10.** A 2.0 kg ball moving to the right at 6.0 m/s catches up to and collides **elastically** with a 4.0 kg ball moving to the right at 1.0 m/s. What is the velocity of the 2.0 kg ball after the collision? Positive to the right / negative to the left.
a) +4.33 m/s b) +2.67 m/s c) +0.67 m/s d) 0 m/s
e) −0.67 m/s



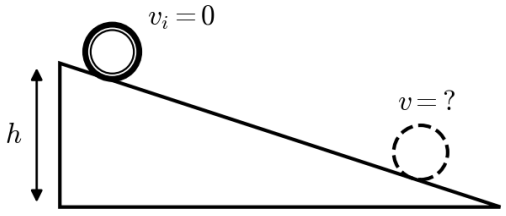

**11.** A 0.010 kg bullet moving at 400 m/s embeds itself in a 2.0 kg block of wood hanging at rest from light strings. How high does the block swing?
a) 0.10 m b) 0.20 m c) 0.40 m d) 1.99 m e) 40.6 m



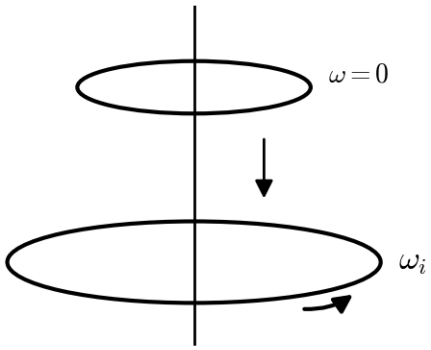

**12.** A 3.0 kg mass hangs from a light string wrapped around a pulley that is a thin ring ($I = MR^2$, $M = 4.0$ kg) on a frictionless axle. The mass is released. What is the tension in the string while the mass falls?
a) 0 N b) 11.8 N c) 16.8 N d) 29.4 N e) 42.0 N




**13.** A thin hoop ($I = MR^2$) starts from rest and rolls without slipping down an incline from a height of 1.5 m. How fast is its center moving at the bottom?
a) 3.83 m/s b) 4.43 m/s c) 4.58 m/s d) 5.42 m/s
e) 7.67 m/s



**14.** A disk ($I = \frac{1}{2}MR^2$, $M = 200$ kg, $R = 1.5$ m) rotates at 4.0 rad/s. A second disk ($I = \frac{1}{2}MR^2$, $M = 100$ kg, $R = 1.0$ m) that is **not** rotating is dropped centered onto it, and the two end up turning together. What is their final angular velocity?
a) 2.00 rad/s b) 2.67 rad/s c) 2.77 rad/s d) 3.27 rad/s
e) 4.00 rad/s




**15.** A merry-go-round (a disk, $I = \frac{1}{2}MR^2$, $M = 200$ kg, $R = 2.0$ m) rotates at 1.0 rad/s. A 30 kg child runs at 4.0 m/s tangent to the rim, **opposite** to the direction the rim is moving, and jumps on at the rim. What is the new rotation rate (in the merry-go-round's original direction)?
a) 0.31 rad/s b) 0.40 rad/s c) 0.77 rad/s d) 1.23 rad/s
e) 1.60 rad/s

